

- [Charts: visualizing data](#)
- [Getting started with BigQuery](#)

Machine Learning

These are a few of the notebooks related to Machine Learning, including Google's online Machine Learning course. See the [full course website](#) for more.

- [Intro to Pandas DataFrame](#)
- [Intro to RAPIDS cuDF to accelerate pandas](#)
- [Getting Started with cuML's accelerator mode](#)

Using Accelerated Hardware

- [Train a CNN to classify handwritten digits on the MNIST dataset using Flax NNX API](#)
- [Train a Vision Transformer \(ViT\) for image classification with JAX](#)
- [Text classification with a transformer language model using JAX](#)

✓ Featured examples

- [Train a miniGPT language model with JAX AI Stack](#)
- [LoRA/QLoRA finetuning for LLM using Tunix](#)
- [Parameter-efficient fine-tuning of Gemma with LoRA and QLoRA](#)
- [Loading Hugging Face Transformers Checkpoints](#)
- [8-bit Integer Quantization in Keras](#)
- [Float8 training and inference with a simple Transformer model](#)
- [Pretraining a Transformer from scratch with KerasHub](#)
- [Simple MNIST convnet](#)
- [Image classification from scratch using Keras 3](#)
- [Image Classification with KerasHub](#)

```
import numpy as np
import matplotlib.pyplot as plt

student_scores = np.array([
    [85, 78, 92, 88],
    [90, 82, 85, 91],
    [76, 89, 80, 84],
    [88, 90, 87, 86]
])

subjects = ["Math", "Science", "English", "History"]

average_scores = np.mean(student_scores, axis=0)

print("Average score for each subject:")
for subject, avg in zip(subjects, average_scores):
    print(f"{subject}: {avg:.2f}")

highest_index = np.argmax(average_scores)

print("\nSubject with the highest average score:")
print(subjects[highest_index])
print(f"Average Score: {average_scores[highest_index]:.2f}")

plt.figure(figsize=(8, 5))

bars = plt.bar(subjects, average_scores)
bars[highest_index].set_color("orange")

for bar in bars:
    height = bar.get_height()
    plt.text(
        bar.get_x() + bar.get_width() / 2,
        height + 0.2,
        f"{height:.2f}",
        ha="center"
    )
```

```
plt.title("Average Scores by Subject")
plt.xlabel("Subjects")
plt.ylabel("Average Score")
plt.ylim(0, 100)
plt.grid(axis="y", linestyle="--", alpha=0.7)

plt.show()
```

Average score for each subject:

Math: 84.75

Science: 84.75

English: 86.00

History: 87.25

Subject with the highest average score:

History

Average Score: 87.25

